

InventOR: A Human-Reviewed, Prompt-Configured Large Language Model Decision Support System for Material-Level Inventory Control

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Abstract

Material planners must set a reorder point, a safety stock and an order quantity for thousands of plant-material pairs, usually under ERP parameters that were last revised long ago. We define and evaluate *InventOR*, a decision support system (DSS) for this recurring decision. For each material, a large language model (LLM) that is configured only through a prompt, without training or fine-tuning, proposes a reorder-point/order-quantity (r, Q) policy with a written justification and an escalation flag. Deterministic software gates the proposal against minimum order quantities and storage limits and replays it on held-out demand before the planner, who remains the decision-maker, reviews it. We derive six design requirements and evaluate the artifact on 365 plant-material pairs from three plants of an industrial manufacturer (346 in the common-feasible cohort), using three independent LLM runs, seven reference arms that differ in information and method, and an ablation that withholds the ERP planning parameters. In a hard-limit storage replay, the LLM policies cost about 11% less than ERP-derived pa-

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rameters at a mean fill rate 0.1–0.4 points lower, but 6%, 8% and 18% more than history-only normal-approximation, Croston/SBA and empirical-quantile (r, Q) baselines. The saving over ERP parameters comes from carrying 13–17% less inventory with 11–20% more stockout days; when shortages are priced at one unit cost, ERP-derived parameters are cheaper than the LLM and a simulation-tuned policy is cheapest. Static storage checks are insufficient: unconstrained trajectories exceed storage limits for 53–66% of pairs. The simulation supports InventOR as a checkable, human-reviewed candidate generator, not as an optimizer that beats data-driven baselines; benefits to planners are untested. We contribute design principles, trust and escalation rules, and a research agenda.

Keywords: Decision Support Systems, Large Language Models, Generative Artificial Intelligence, Inventory Control, Human–AI Decision Making, Design Science, Operations Research

1. Introduction

Replenishment parameters in manufacturing are set per plant-material pair: a reorder point that triggers an order, a safety stock that absorbs demand and lead-time variability, and an order quantity. In large ERP systems these values often remain at levels set years earlier, while demand patterns, supplier performance and storage constraints change. Revisiting them materially requires analysts who can fit demand and lead-time statistics, apply inventory theory, respect minimum order quantities (MOQs) and storage limits, and explain the result to planners. That scarce skill is exactly what a conversational large language model (LLM) with a code interpreter appears to offer, and a rapidly growing literature explores such tools for supply-chain and operations decisions [1, 2, 3, 4, 5].

13 The promise is real, but so are three problems that limit its use in prac-
14 tice. First, LLM outputs can be unfaithful or fabricated [6], so a proposed
15 policy must be treated as a candidate and not a decision. Second, claims of
16 superiority over ERP parameters are hard to interpret when the LLM and
17 the comparators are given different information; an LLM that sees a plant’s
18 ERP reference values and is compared with an OR policy that does not, or
19 the reverse, tells us little about the LLM. Third, inventory policies live inside
20 physical limits: a storage rule that holds when the policy is written down
21 can be violated as stock builds up over time.

22 This paper treats the problem as one of decision support system (DSS)
23 design. We define *InventOR* as a human-reviewed, model-driven DSS for
24 the recurring decision of setting replenishment parameters per plant-material
25 pair (Section 2). A planner remains the decision-maker. The system supplies,
26 for each material, a candidate (r, Q) policy proposed by an enterprise LLM
27 application that is configured only through a prompt and receives no project-
28 specific training or fine-tuning, together with a justification, an escalation
29 flag and a counterfactual simulation; deterministic software parses, gates
30 and simulates every proposal before the planner reviews it and alone decides
31 whether any ERP parameter is changed. The pattern is not specific to
32 inventory: any production-planning decision in which parameters are set per
33 item and refreshed periodically (lot sizes, planned lead times, buffer levels,
34 dispatching rules) has the same structure of many semi-structured decisions,
35 scarce analyst time, hard constraints and a need for accountable evidence.
36 The design lessons below therefore concern LLM-based decision support for
37 production and logistics planning in general, and not only inventory control.

38 We follow a design-science logic [7, 8]: we state the problem and its scope,
39 derive design requirements, build the artifact, and evaluate it against those

40 requirements on real industrial data. Four research questions structure the
41 paper.

42 **RQ1. *Design.*** How should an LLM be embedded in a DSS so that its
43 proposals are bounded, verifiable, reproducible and reviewable by a
44 planner?

45 **RQ2. *Efficacy.*** What cost and service do the proposed policies achieve
46 relative to ERP-derived parameters and to OR policies, when the
47 information available to each approach is controlled?

48 **RQ3. *Feasibility and reliability.*** Do the policies respect physical storage
49 limits along simulated inventory trajectories, and how stable are they
50 across independent runs?

51 **RQ4. *Use.*** Under which conditions should a planner trust, escalate or reject
52 a proposal?

53 Our contributions are:

54 **C1. A definition and scope of the DSS** (Section 2): the decision,
55 the decision-maker, the system boundary, the level of automation and
56 six design requirements that make the role of the LLM explicit and
57 bounded.

58 **C2. A DSS architecture and operating rules** (Section 4) that separate
59 generation, deterministic gating, simulation and planner review, with
60 explicit rules for when output is trusted, escalated or rejected, and
61 design principles derived from this application (Section 8).

62 **C3. An information-controlled, trajectory-level evaluation** (Sec-
63 tion 5) with seven reference arms that differ in the information they use

and in method (ERP-derived, ERP-informed, normal-approximation, Croston/SBA, empirical-quantile and simulation-tuned policies), an LLM ablation denied the ERP planning parameters, and a hard-limit storage replay that is the primary result. This separates what the LLM adds from what it is given.

C4. Complete-cohort industrial evidence from three LLM runs on 365 plant-material pairs, with plant-level, regime-level and sensitivity analyses, and managerial and research implications.

Our main findings are as follows. Against ERP-derived parameters the LLM policies are about 11% cheaper at a mean fill rate 0.1–0.4 points lower, but the LLM costs 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile (r, Q) baselines at essentially equal service, and its saving over ERP comes from carrying less inventory at the price of more stockout days. Storage limits that hold in a static check are violated along unconstrained trajectories for 53–66% of pairs. The simulation therefore does not show superior optimization by the LLM; it shows that a prompt-configured LLM can produce checkable candidate policies without code or training, whose benefit to planners remains to be tested.

2. Problem, Scope and Definition

2.1. The decision problem

The decision supported is the periodic (re)setting of three replenishment parameters, a reorder point, a safety stock and an order quantity, for each plant-material pair held in an ERP system. In the classification of Gorry and Scott Morton [9] it is an operational-control decision that is partly structured: the calculation is textbook, but the choices of estimator, the treat-

ment of intermittent or trending demand, the plausibility of recorded lead times and the handling of constraints require judgement that is applied inconsistently when thousands of materials compete for a few analysts' time. The decision-maker is the material planner or inventory analyst; the decision is repeated, high-volume and consequential for working capital, service and storage. Hard constraints are the minimum order quantity (MOQ) and a physical storage limit. The feedback loop is slow: a changed parameter shows its effect only in subsequent demand.

2.2. Definition and level of automation

Definition. *InventOR is a human-reviewed, model-driven decision support system that, for one plant-material pair at a time, (i) obtains from a large language model, configured only by a written prompt, a candidate (r, Q) policy with a justification and an escalation flag; (ii) verifies the candidate deterministically against explicit constraints; (iii) evaluates it by counterfactual simulation against reference policies on held-out demand; and (iv) presents the result to a planner, who alone decides whether ERP parameters are changed.*

In the classical DSS architecture of data, model and dialog components [10, 11], the LLM and the simulator form the model component, the material-scoped data extract and the prompt form the data and knowledge components, and the review interface is the dialog component (Section 4.1). In terms of the levels of automation of Parasuraman et al. [12], InventOR operates at a low-to-intermediate level: it selects and suggests one alternative and leaves execution to the human. This is a design decision, not a limitation: acceptance of automated suggestions is accompanied by complacency

and automation bias [13], which the review rules of Section 4.3 are meant to counter.

2.3. Scope

Table 1 states what is and is not part of the system and of the evaluation, so that the claims of the paper are read at the right level.

Table 1: Scope of InventOR and of its evaluation.

Dimension	In scope	Out of scope
Decision	Reorder point, safety stock and order quantity per plant-material pair, reviewed periodically	Order release, supplier selection, pricing, forecasting-model development
Inventory system	Single item, single echelon, continuous review, lost sales, (r, Q) family	Multi-item coupling, shared capacity, multi-echelon networks
Inputs	Daily demand and realized lead times, working-day flags, cost and constraint parameters, ERP reference parameters	External signals, supplier forecasts, market data
Role of the LLM	Candidate generator with justification and escalation flag	Forecasting, order placement, running the simulation, final decision
Human role	Planner reviews and decides; rules for trust, escalation and rejection	Autonomous execution without review
Evaluation	Ex-post simulation on real post-cut-off demand, information-controlled comparison, hard storage-limit replay, repeated runs	Field trial, planner workload or decision quality, user acceptance

2.4. Design requirements

Four sources motivate the requirements: the known unreliability of LLM outputs [6], guidance on human–AI interaction and accountability [14, 15], the run-to-run nondeterminism of LLM inference [16], and the practical constraints of inventory control. They yield six requirements (Table 2), each tied to a design feature (Section 4) and an evaluation (Section 6).

2.5. Research approach

We follow the design-science research process of problem identification, requirements, artifact design and evaluation [7, 8]. In the evaluation frame-

Table 2: Design requirements, the design features that address them and where they are evaluated.

ID	Requirement	Design feature	Evaluation
DR1	Bounded generation: the LLM proposes and has no authority	Narrow candidate-generator role; prohibited methods; no order placement	Sections 6.1, 6.3
DR2	Verifiable feasibility along time, not only at a point	Deterministic gate; hard-limit trajectory replay	Sections 6.1, 6.5
DR3	Counterfactual evidence for the reviewer	Held-out simulation against reference arms	Sections 6.2, 6.8
DR4	Information transparency: comparisons control what each arm sees	Seven reference arms; LLM ablation without ERP parameters	Sections 6.3, 6.4
DR5	Reproducibility and accountability of the configuration	Frozen prompt file; per-call log; repeated runs	Section 6.7
DR6	Human control with explicit escalation	Trust, escalate and reject rules; escalation flag	Sections 4.3, 6.6

work of Venable et al. [17], our evaluation is artificial and ex post: it uses real industrial data and a faithful simulation but no live users. It therefore establishes technical efficacy, feasibility and reliability; it does not establish usability, adoption or improved planner decisions, which we list as limitations and research directions.

3. Related Work and Positioning

Decision support systems and human-AI decision making. DSS research has long distinguished data-, model- and knowledge-driven systems and has stressed that the value of a DSS lies in improving the judgement of a human decision-maker rather than in replacing it [9, 10, 11]. LLMs add a new kind of model component: it is configured through language, produces free-text justification and is not deterministic. Research on human-automation interaction shows that the way suggestions are scoped, presented and verified determines whether users rely on them appropriately or over-rely on them [12, 13], and guidelines for human-AI interaction call for explicit scoping, explanation and correction mechanisms [14]. Our contribution to this stream is

145 a concrete, evaluated instance of an LLM-based DSS for an operational pro-
 146 duction decision, with design requirements that are tied to evidence. Recent
 147 DSS work derives design principles for LLM-based assistants [18], reports de-
 148 sign and pilot implementations of AI-enabled decision support in operations
 149 [19], and shows that the uptake of algorithmic advice depends on trans-
 150 parency and on the user’s metacognition [20, 21]; evaluations of data-driven
 151 forecasting systems in practice likewise stress lessons beyond accuracy [22].
 152 Model-driven decision support is also known to shape how groups handle
 153 disagreement [23]. Human override of algorithmic safety-stock advice is a
 154 documented risk in inventory settings [24], which is why InventOR keeps the
 155 planner as decision-maker and makes the evidence for each proposal explicit.

156 *LLMs in operations and supply-chain decisions..* Reviews and frameworks
 157 map the opportunities for generative AI in supply chains [1, 2, 25, 26, 3, 4, 27]
 158 and stress that adoption depends on assigned human roles, data quality and
 159 governance [5, 28, 29, 30, 31]. Case evidence from manufacturing planning
 160 is still thin: examples include a fine-tuned LLM for digital-manufacturing
 161 optimization [32] and a transformer trained to set order timing and quan-
 162 tity, benchmarked against (r, Q) [33]; experiments with managers also show
 163 that ChatGPT reproduces human decision biases [34]. A recent conceptual
 164 line proposes agentic digital twins that combine AI agents with model-based
 165 simulation and optimization to support decisions [35], and an LLM-plus-
 166 digital-twin method has been built for closed-loop supply-chain operations
 167 [36]. Our decision support system follows the same logic at a smaller scale:
 168 an LLM proposes and a simulation model checks. Most technical LLM work
 169 focuses on translating natural-language problem statements into optimiza-
 170 tion models and solver code [37, 38, 39, 40], on multi-agent control and

171 consensus-seeking in inventory flows [41, 42], on extracting supply-network
 172 structure from text [43, 44], on disruption modelling [45], or on evaluating
 173 LLMs on classical operations-research tasks such as vehicle routing [46]. The
 174 closest LLM study to ours uses an LLM as a forecasting copilot for spare-
 175 parts demand, combined with simulation, and reports high variability for
 176 irregular demand that can be reduced by averaging repeated queries and en-
 177 riching prompts [47]. We share the concern about run-to-run variability and
 178 test it directly (Section 6.7), but differ in that the LLM proposes a complete
 179 replenishment policy rather than a forecast.

180 *Data-driven inventory control without LLMs.* Learning-based replen-
 181 ishment is well established, including data-driven policies learned from
 182 sequentially observed non-stationary demand [48] and forecasts augmented
 183 with leading indicators [49], and heuristics remain competitive in constrained
 184 settings such as perishable inventory [50]. Deep reinforcement learning has
 185 been applied to omnichannel and multi-echelon systems [51, 52, 53] and
 186 aerospace replenishment [54]; deep-learning forecasting has been integrated
 187 with replenishment [55]; contextual distributionally robust optimization
 188 uses side information to set order quantities [56]; and distribution-free
 189 online algorithms handle capacitated warehouses under uncertain price
 190 and demand [57]. Robust inventory control under demand and lead-time
 191 uncertainty has a longer history [58, 59, 60], and lead-time variability and
 192 dependence are known to propagate through ordering decisions [61], while
 193 safety stock and safety lead time are alternative buffers against unreliability
 194 [62]. The closest operational neighbour is a trained data-driven framework
 195 that jointly sets replenishment parameters, safety stock and safety lead
 196 time on real automotive data [63]. Reinforcement learning has also been

combined with demand forecasting to optimize order points [64]. These methods require model training, historical scenario generation or tuned hyperparameters for each setting; InventOR requires none, at the price of an LLM-based generator whose internal logic cannot be inspected beyond its written justification.

Intermittent and spare-parts demand. Material histories in manufacturing are frequently intermittent or lumpy. The statistical literature offers specialized estimators and classification schemes [65, 66, 67, 68, 69], and recent studies develop feature-based forecast combinations and temporal aggregation for intermittent demand [70, 71], forecasting for spare parts across product life-cycle phases [72], and integrated forecasting and planning or multi-echelon stocking models for repairable and spare parts [73, 74]. Related industrial studies examine how manufacturers identify and select safety buffers [75]. Our regime analysis (Section 6.6) shows where an LLM that is forbidden from using specialized intermittent-demand estimators behaves differently on lumpy materials.

Table 3 summarizes these streams and states how the contribution of InventOR differs. InventOR is not a new inventory model and does not claim to beat trained or optimized methods. It contributes (i) a DSS design in which an off-the-shelf LLM is a bounded candidate generator inside a reviewed process, (ii) an evaluation that controls the information available to each arm, and (iii) industrial evidence on cost, service and storage feasibility. The (r, Q) policy it uses is the standard workhorse of continuous-review control [76, 77, 78].

Table 3: Literature positioning of InventOR relative to recent work on AI-supported inventory and supply-chain decisions.

Study	Approach	Decision	Relation to InventOR
<i>LLMs for operations and supply chains</i>			
Jackson et al. [1]; Fosso Wamba et al. [2]	Framework / review	Broad supply-chain use	Motivates LLM use; no inventory-policy evidence
Song et al. [3]; Panja et al. [4]	Systematic reviews	LLMs in SCM and logistics	Place InventOR in the emerging empirical gap
Trifone et al. [5]; Cannas et al. [28]; Venkatesh et al. [29]	Adoption studies, search agenda	AI roles, barriers, data quality	Support human-in-the-loop design and review rules
Ivanov [35]; Le et al. [36]	Agentic LLM digital twins	Model-based decision support	Pair AI proposals with simulation; InventOR applies this to (r, Q) parameters
Nakhost et al. [47]	LLM + simulation	Spare-parts forecast	Closest LLM study; forecast, not policy
Jannelli et al. [42]; Quan and Liu [41]	Multi-agent LLM	Sequential ordering, consensus	Per-period agent decisions in simulation
AlMahri et al. [43]; Zheng and Brintrup [44]	LLM + knowledge graphs	Supply-network visibility	LLM for information extraction, not inventory control
Hamdan and Brahimi [46]	LLM benchmark	Routing	Evaluates LLM on a classical OR task
Li et al. [37]; Huang et al. [38]; Ahmadi-teshnizi et al. [39]; Zhang et al. [40]	NL→model solver	General OR models	Model formulation; not a fixed policy family
<i>Data-driven inventory control without LLMs</i>			
Gabellini et al. [63]	Trained data-driven	Replenishment, safety stock and safety lead time	Closest OR neighbour; needs training; automotive data
Kolyaei et al. [51]; Stranieri et al. [52]; Wang et al. [54]	Deep reinforcement learning	Replenishment	Learned policies; require training environments
Ma [55]; Wang et al. [56]; Zhang et al. [57]	Deep learning / contextual DRO / online algorithm	Forecast-plus-order; capacitated replenishment	Training, tuning or worst-case design per setting
<i>Intermittent and spare-parts demand</i>			
Sanguri et al. [70]; Li et al. [71]; Goldsmith and Sachs [72]	Statistical ML forecasting	Intermittent demand	Specialized estimators the LLM is not allowed to use
Babaveisi et al. [73]; Caserta and D'Angelo [74]	Optimization	Spare-part stocking	Multi-echelon or repairable settings; outside our single-echelon scope
Prak et al. [68]; Babai et al. [69]	Estimation review	Inventory under intermittency	Regime-aware benchmarks for future work
InventOR (this paper)	(this Prompted LLM + deterministic gate and simulation)	(r, Q) per plant-material pair	No training; reviewed; information-controlled evaluation

221 4. The InventOR Decision Support System

222 4.1. Architecture and components

223 Table 4 maps the classical DSS components [10, 11] to their realization in
 224 InventOR and to the design requirements of Section 2.4. The generator is the
 225 only component whose internal logic is not inspectable, and it is surrounded
 226 by components that are deterministic and auditable. In the evaluated proto-
 227 type the dialog component is the per-pair output record and the comparison
 228 tables; an interactive planner interface was not built or evaluated.

Table 4: DSS components of InventOR.

Component	Realization in InventOR	Requirement
Data	Material-scoped extract of daily demand, lead times, working-day flags, cost and constraint parameters, and ERP reference values (Table 5); the set of fields supplied is controlled and logged	DR4
Knowledge	System prompt file defining the task, the prohibited methods, the storage rules and the output schema (Supplementary File S1)	DR1, DR5
Model: generator	Enterprise LLM application with a code interpreter, configured only by the prompt; proposes one (r, Q) policy, a justification and an escalation flag	DR1
Model: verifier and simulator	Deterministic parser and gate; held-out simulation with hard storage-limit replay; reference arms	DR2, DR3
Dialog (review)	Policy, justification, flag and simulated comparison presented to the planner under the trust, escalate and reject rules	DR6
Governance	Frozen prompt file, per-call log of input hash, time, session and raw response, repeated runs	DR5

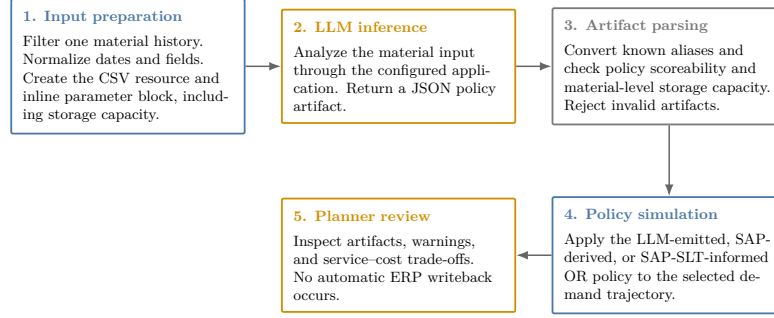
229 4.2. Operation in four steps

230 A planner’s question is: *given what this material has done in the past,*
 231 *how much safety stock should I hold, at what stock level should I reorder,*
 232 *and how much should I order?* InventOR answers this question in four steps
 233 (Figure 1).

234 **Step 1. Generate.** The LLM, accessed through an enterprise application
 235 with a code interpreter, receives one material’s history (daily de-

InventOR prototype workflow

Executed material-level path from prepared history to planner-reviewable simulation output.



Deterministic components reproduce simulation results from stored artifacts; LLM inference itself may vary.

Figure 1: InventOR workflow: material data are prepared, an LLM proposes a policy, deterministic software parses, checks and simulates it, and a planner reviews the result.

mand, realized lead times, working-day flags) and a short parameter block (unit cost, holding rate, ordering cost, MOQ, service target, storage limit, and current ERP safety stock and safety lead time for comparison). A written prompt (the only configuration; no weights are changed) [79] instructs it to compute demand and lead-time statistics, classify the demand pattern, apply an economic-order-quantity (EOQ) argument and a normal-approximation safety stock, respect the storage limit, and report a structured policy together with a short justification and escalation flag.

Step 2. Gate. Deterministic code extracts the policy, rejects malformed or infeasible outputs (missing or non-numeric fields, non-positive reorder point or order quantity, negative safety stock, reorder point below safety stock, safety stock plus order quantity above the storage limit) and retries the generation up to three times. The backtest applies one further admission check, that the order quantity is not

252 below the MOQ.

253 **Step 3. Simulate.** Each accepted policy is replayed on the material’s de-
254 mand observed *after* the data cutoff, a held-out evaluation in the
255 sense of out-of-sample forecast testing [80, 81], together with the
256 comparator policies, producing holding and ordering cost, fill rate
257 and stockout days. This plays the role that model-based simulation
258 plays in digital-twin decision support [35], at the scale of a single
259 policy.

260 **Step 4. Review.** A planner sees the policy, the LLM’s justification and
261 flag, and the simulated comparison. Only the planner changes ERP
262 parameters.

263 The role of the LLM is deliberately narrow. It is a *candidate generator*
264 whose output is never trusted without the gate, and it is not used to fore-
265 cast, to place orders, or to run the simulation. The prompt prohibits Monte
266 Carlo simulation, distribution fitting, machine-learning libraries and special-
267 ized intermittent-demand estimators, so that the policy rests on transparent
268 statistics a planner can reproduce.

269 4.3. Trust, review and reject rules

270 We propose three explicit rules for planners, in line with guidelines for
271 human–AI interaction and with internal-auditing practice for AI systems
272 [14, 15], and motivated by the evidence in Section 6. The rules and the 0.5-
273 point threshold are design proposals that we did not evaluate with planners;
274 with the baselines used here the trust rule would be met by only a minority
275 of pairs (the LLM is cheaper than the history-only OR in 74–77 of 346),
276 which is itself a finding about the LLM.

277 **Trust (apply after review)** when the proposal passes the gate, the sim-
278 ulated trajectory respects the storage limit when orders are truncated at it,
279 and the simulated cost is not above that of the best history-only OR ref-
280 erence (Croston/SBA or empirical quantile) with a fill rate not more than
281 about 0.5 points lower.

282 **Escalate** when the LLM raises an escalation flag, the material is lumpy or
283 has strongly seasonal or trending demand, the plant or calendar data were
284 repaired, or the policy differs strongly between repeated generations.

285 **Reject** when the proposal fails the gate after the retry budget, exceeds the
286 storage limit in the hard-limit replay, or is dominated (higher cost and lower
287 fill) by a history-only reference.

288 *4.4. Provenance and reproducibility of the configuration*

289 The system prompt, which fully defines the task, the forbidden methods,
290 the capacity rules and the output schema, is a plain-text file that is deployed
291 unchanged in the enterprise application; it is distributed with the paper
292 (Supplementary File S1) and the code repository. Every call logs the SHA-
293 256 of its input data, the UTC time, the application session identifier and
294 the full raw response. The platform does not expose the underlying model
295 identifier or sampling settings to the client; the evaluated model (Anthropic
296 Claude Sonnet 4.6, with a code interpreter) is therefore declared by the
297 operator and is not machine-verifiable. We report this limitation openly and
298 base the reproducibility argument on the prompt file, the per-call logs and
299 three independent runs (Section 6.7) rather than on a model checkpoint.

300 5. Application and Evaluation Design

301 5.1. Evaluation approach

302 The evaluation addresses each design requirement in turn (Table 2):
303 whether the pipeline returns valid, feasible policies (DR1, DR2); what cost
304 and service the policies achieve against reference arms on held-out demand
305 (DR3); how much of that result is due to the information given to the LLM
306 rather than to its reasoning (DR4); whether storage limits hold along the
307 simulated trajectory (DR2); how stable policies are across independent runs
308 (DR5); and where proposals should be escalated (DR6). The primary basis
309 is the hard-limit replay with working-day receipts and no shortage penalty;
310 sensitivity analyses vary these choices. The evaluation is ex post and uses
311 no live users (Section 2.5).

312 5.2. Industrial application and data

313 The application is a pilot at a large manufacturer. The dataset contains
314 233,078 daily records for 411 plant-material pairs from three plants (A, B
315 and C), with a data cutoff on 10 March 2026 and a post-cutoff horizon
316 observed to 27 August 2026. Of the 411 pairs, 365 are eligible for scoring
317 (234 in Plant A, 65 in Plant B, 66 in Plant C) and are sent to the LLM. For
318 19 pairs (all in Plant A) the supplied MOQ plus the reference safety stock
319 exceeds the storage limit, so no comparator can satisfy both rules; these are
320 excluded from outcome comparison, leaving a *common-feasible cohort* of 346
321 pairs (215, 65 and 66). All 365 pairs receive an LLM policy. Demand is
322 concentrated: the five largest materials account for 40.3% of demand and
323 the ten largest for 57.8%. The source working-day flag of Plant C is all zero;
324 we reconstruct it from the same-date majority of Plants A and B (ties count

as working days) and test the effect of this repair in Section 6.8. Such data-quality problems are a known threat to analytics-based decision support [82] and, for inventory systems, to record accuracy [83, 84, 85].

Table 5: Fields of the material-scoped CSV given to the LLM.

Column	Type	Description and use
date	date	Observation date; time ordering and aggregation
actual_demand	numeric	Realized daily demand; mean, variability, intermittency, regime
delivery_time	numeric	Recorded replenishment lead time (receipt rows)
is_working_day	binary	Working-day flag; demand statistics over working days
minimum_order_quantity	numeric	MOQ; order-size constraint
unit_cost,	numeric	Unit value, annual carrying-cost rate, fixed ordering cost;
holding_cost_rate,		EOQ and cost evaluation
ordering_cost		
service_level_target	numeric	Target service used in safety-stock computation
max_storage_units	numeric	Storage limit: $MOQ \leq Q$ and $SS + Q \leq \text{max_storage_units}$
safety_stock_units,	numeric	ERP planning parameters (current safety stock, safety lead time); withheld in the ablation arm
safety_time_workdays_actual		
reliability_calc,	mixed	Supplier and lane context
supplier_plant_distance,		
transportation_network		

5.3. Reference arms and information content

The LLM and the classical comparators do not necessarily use the same information, and this matters for the interpretation of any cost difference, because ERP safety stock and safety lead time embed accumulated planner judgement, and safety stock and safety lead time are alternative ways of buffering unreliability [62]. Planned lead times can also drift upward when they are adjusted dynamically (the “lead time syndrome” [86]), which would inflate any comparator that adds them to the lead time. We therefore construct arms that make the information explicit (Table 6).

Three further arms use no ERP input and represent stronger data-driven practice than the history-only OR. The *Croston/SBA OR* replaces the mean daily demand by the Croston estimate with the Syntetos–Boylan bias correction [65, 66] (smoothing constant 0.1, working-day demand history), uses

Table 6: Comparison arms and the information each uses. All arms respect MOQ and are projected to the storage limit when possible.

Arm	Parameters from	ERP used	inputs	Purpose
SAP-derived	ERP safety stock; EOQ order quantity	Safety stock		ERP-derived reference (not observed practice)
SLT-informed OR	Stationary (r, Q) with ERP safety lead time added to the lead time	Safety lead time		OR policy using the ERP safety lead time
ERP-floor OR	SLT-informed OR with safety stock floored at the ERP value	Both		OR arm matched to the LLM’s full information
History-only OR	Stationary (r, Q) from demand and lead-time history, mean lead time	None		ERP-independent baseline
Croston/SBA OR	(r, Q) with the Syntetos–Boylan-approximated Croston demand rate and its forecast-error spread, mean lead time	None		Intermittent-demand baseline
Empirical-quantile OR	(r, Q) with the reorder point set to the service-level quantile of historical rolling lead-time demand	None		Distribution-free baseline
Simulation-tuned (r, Q)	Reserve factor and order-quantity multiple chosen by simulation on the last 250 working days before the cutoff	None		Trained data-driven baseline
LLM (runs 1–3)	LLM proposal with all fields of Table 5	Both		Main treatment
LLM without ERP	LLM with ERP safety stock and safety lead time set to “not supplied”	None		Ablation: what the LLM does when it cannot anchor on ERP

341 the root mean squared one-step forecast error as demand spread and the
 342 history-only lead-time moments, which suits intermittent demand [67]. The
 343 *empirical-quantile OR* sets the reorder point to the service-level quantile
 344 of the historical lead-time demand, computed from rolling sums over the
 345 rounded mean lead time in working days, and needs no distributional as-
 346 sumption. The *simulation-tuned (r, Q)* policy is a learned baseline: from a
 347 grid of reserve factors and order-quantity multiples around the history-only
 348 policy it selects the cheapest candidate whose fill rate on the last 250 work-

ing days *before the cutoff* meets the service target (or the best-fill candidate if none does). No post-cutoff data are used in any of these arms. We do not implement a deep reinforcement-learning baseline [52, 51]; the simulation-tuned policy is the trained comparator, and Section 8 lists neural baselines as a limitation.

5.4. Simulation, capacity replay and metrics

Each pair is simulated independently on the post-cutoff horizon as a single-item, single-echelon continuous-review (r, Q) system [76, 87, 88, 78, 77] with lost sales [89], in the spirit of simulation-based experiments on supply and timing uncertainty [90], a shared opening stock, an empty pipeline and arrivals after the rounded mean lead time in working days (Appendix Appendix A). MOQs are treated as hard order-size constraints, as imposed by suppliers [91], and storage limits as capacity bounds on the replenishment decision [57]. The primary metrics are the mean material-horizon holding-plus-ordering cost, the mean material-level fill rate, the aggregate (volume-weighted) fill rate and the number of stockout days. We report paired differences with 10,000-sample bootstrap 95% intervals.

A static check $SS + Q \leq \text{max_storage_units}$ does not show that storage limits are respected as the inventory evolves, because the reorder point includes lead-time demand and inventory on order can exceed the limit when it arrives. We therefore record, for every policy and every simulated day, the post-receipt on-hand stock and report the share of pairs and days above the limit (the *unconstrained* simulation). We then *replay* every policy with each order truncated to the storage limit minus the current inventory position (the *hard-limit replay*). The hard-limit replay is our primary evaluation.

374 5.5. *Repeated runs and sensitivities*

375 To examine run-to-run variation, the entire cohort is generated three
376 times (runs 1, 2 and 3) with the same prompt file and the same data. Run 1
377 serves as the reference for tabulated results unless stated otherwise; runs 2
378 and 3 provide replicates. Sensitivity analyses vary the working-day recon-
379 struction (majority, any-plant, strict-all, exclusion of Plant C and calendar-
380 day arrivals), the shortage penalty (zero, one and five times unit cost).

381 6. Results

382 6.1. *Completion and policy characteristics*

383 All four runs (three with the full input and one ablation, Section 6.4)
384 return a valid, storage-feasible (r, Q) policy for all 365 pairs after at most
385 three attempts per pair. The validity gate rejects non-positive reorder points
386 and order quantities, negative safety stocks and reorder points below the
387 safety stock. The ERP safety stock is zero for 198 of the 365 pairs (188 of
388 the 346 in the cohort below), so for a majority of pairs the SAP-derived arm
389 holds no safety stock at all.

390 6.2. *Headline comparison with ERP-derived parameters*

391 Unless stated otherwise, all results from here on use the hard-limit replay
392 (every order truncated at the storage limit) on the 346-pair common-feasible
393 cohort, with working-day receipts and no shortage penalty; Section 6.5 re-
394 ports the unconstrained simulation for comparison. Table 7 compares all
395 arms, including the LLM cost relative to each arm in the three runs (posi-
396 tive values mean that the LLM costs more). In run 1 the LLM policies cost
397 11.4% less than SAP-derived policies and 30.5% less than the SLT-informed

OR comparator (mean 2,928 versus 3,305 and 4,211). The mean fill rate is 0.18 points lower than for SAP-derived (98.11% versus 98.30%; bootstrap interval -0.54 to $+0.10$) and 0.11 points lower than for the SLT-informed OR. The mean cost difference against SAP-derived is -377 (95% interval -651 to -154); the LLM is cheaper than SAP-derived in 241 of 346 pairs and, in 226 pairs, simultaneously no more expensive and no worse in fill. The LLM incurs more stockout days in total (910 versus 804 and 831), and aggregate fill is 78.87% versus 79.34% and 79.15%.

The cost decomposition shows where the saving comes from. Relative to SAP-derived parameters the LLM carries about 17% less average inventory (16,256 versus 19,565 units) and has about 16% lower holding cost, while it places about the same number of orders per year (17.3 versus 17.6). The saving is therefore a trade of inventory for service: the LLM holds less stock and accepts 11–20% more stockout days (runs 3 and 2). It is not the consequence of a better ordering rhythm, and it is not free of service cost. Section 6.8 shows that, once shortages carry a cost, this trade no longer favours the LLM.

Table 7: Hard-limit replay, common-feasible cohort ($n = 346$), working-day receipts, zero shortage penalty. Cost is the mean material-horizon holding-plus-ordering cost; average inventory is in units; orders are per year. The last column gives the LLM cost relative to the arm in runs 1/2/3 (positive: the LLM costs more).

Arm	Mean cost	Holding	Ordering	Avg. inv.	Orders	Mean fill (%)	Aggr. fill (%)	Stockout days	LLM vs arm (%)
ERP-derived (SAP)	3,305	2,172	1,132	19,565	17.6	98.30	79.34	804	$-11.4 / -11.3 / -11.4$
SLT-informed OR	4,211	2,608	1,603	21,614	24.5	98.23	79.15	831	$-30.5 / -30.4 / -30.5$
ERP-floor OR	4,248	2,633	1,616	21,937	24.7	98.23	79.15	831	$-31.1 / -31.0 / -31.1$
History-only OR	2,764	1,725	1,039	15,972	16.3	98.14	78.72	874	$+5.9 / +6.0 / +5.9$
Croston/SBA OR	2,705	1,665	1,040	15,521	16.3	98.30	78.87	803	$+8.2 / +8.4 / +8.3$
Empirical-quantile OR	2,489	1,543	946	13,331	14.8	98.35	78.87	747	$+17.7 / +17.8 / +17.7$
Simulation-tuned (r, Q)	3,266	2,473	793	11,914	12.6	96.05	84.84	1,973	$-10.4 / -10.3 / -10.3^\dagger$
LLM (run 1)	2,928	1,824	1,104	16,256	17.3	98.11	78.87	910	
LLM (run 2)	2,931	1,850	1,081	16,599	17.0	97.89	78.70	968	
LLM (run 3)	2,929	1,841	1,088	16,958	17.0	98.16	78.90	890	
LLM without ERP	3,014	1,902	1,112	16,507	17.3	98.06	78.89	890	$-8.8^\ddagger$ (vs SAP)

[†] Other comparisons of the ablation are in Section 6.4. [‡] Different operating point (mean fill 96.05%, aggregate fill 84.8%); the paired interval of the cost difference includes zero, and with priced shortages this policy is the cheapest arm (Section 6.8).

415 *6.3. Information asymmetry: what the LLM adds beyond the ERP inputs*

416 The comparison below uses Table 7. Three results stand out.

417 First, the SLT-informed OR and its ERP-floor variant are the most ex-
 418 pensive arms (4,211 and 4,248). The ERP safety lead time inflates their
 419 effective lead time, hence their buffers; against them the LLM is about 30–
 420 31% cheaper in every run. This comparison therefore mainly measures how
 421 heavily ERP inputs over-buffer, not the quality of the LLM.

422 Second, the *history-only OR*, which uses no ERP inputs, has a cost of
 423 2,764 at a fill of 98.14%. Against it the LLM costs 5.9–6.0% more in all
 424 three runs (run 1: mean difference +164, 95% interval +91 to +240), with
 425 fill differences of -0.03 , -0.25 and $+0.01$ points whose intervals include zero
 426 in all three runs. The LLM is cheaper than the history-only OR in only 74–
 427 77 of 346 pairs. The stronger data-driven arms widen the gap. Against the
 428 Croston/SBA OR (2,705, fill 98.30%) the LLM costs 8.2–8.4% more (run 1:
 429 mean difference +223, interval +97 to +362; the LLM is cheaper in 91–99
 430 pairs), and against the empirical-quantile OR (2,489, fill 98.35%) it costs
 431 17.7–17.8% more (mean difference +439, interval +329 to +557; the LLM
 432 is cheaper in 44–49 pairs) at a mean fill that is 0.24 points lower. Part of
 433 the LLM’s lower cost relative to ERP-derived parameters therefore reflects
 434 the weakness of those benchmarks; the LLM does not match a data-driven
 435 (r, Q) baseline.

436 The simulation-tuned (r, Q) policy sits at a different operating point. It
 437 reaches an aggregate fill of 84.8% with the lowest average inventory (11,914
 438 units) but a lower mean fill (96.05%) and many more, smaller stockouts
 439 (1,973 stockout days), because its tuning criterion is the fill of the last pre-
 440 cutoff year, not the post-cutoff horizon. Against it the LLM costs 10.3–10.4%
 441 less in the zero-penalty cost, with a 2.1-point higher mean fill, but the paired

interval of the cost difference includes zero ($-2,527$ to $+857$ in run 1). Once shortages are priced it becomes by far the cheapest arm (Section 6.8), so its low zero-penalty ranking is not a sign of a weak baseline.

Third, the LLM is nonetheless cheaper than ERP-derived parameters (11.3–11.4%) at a mean fill that is lower by 0.14–0.40 points (the interval excludes zero only in run 2), so it improves on the ERP-derived reference. It does not reach the best classical baselines, which also yield a policy for every material at lower cost; the evidence therefore does not establish a cost advantage of the LLM over data-driven alternatives. Whether the LLM relies on the ERP planning parameters it is given is tested in the ablation of Section 6.4.

6.4. LLM without ERP planning parameters (ablation)

In this ablation the LLM receives exactly the same material histories, cost parameters, MOQ, service target and storage limit as in runs 1–3, but the ERP safety stock and ERP safety lead time are blanked in the uploaded CSV and marked “not supplied” in the inline parameter block; the system prompt file is unchanged. It tests whether the LLM anchors on the ERP planning parameters it is given, and therefore whether the comparison with the ERP-informed arms is confounded by information. All 365 pairs returned a valid, storage-compliant policy.

Policy values. Withholding the ERP parameters changes the LLM’s policies only slightly beyond its ordinary run-to-run variation. The median ratios of safety stock and reorder point to run 1 are both 1.00 (346 common pairs). The reorder point differs by more than 10% in 54.6% of pairs, against 48.3% between runs 1 and 3 and 48.0% between runs 2 and 3 with the ERP parameters supplied; the safety stock differs by more than 10% in 11.0% of

468 pairs, against 9.8% and 9.5%. The median absolute reorder-point difference
 469 is 128 units, against 90 and 84 between ERP-supplied runs. The mean ratio
 470 is slightly above one (1.04 for safety stock, 1.05 for the reorder point), so the
 471 LLM tends to buffer a little more when it has no ERP anchor.

472 *Outcomes.* In the hard-limit replay the LLM without ERP parameters
 473 has a mean cost of 3,014 at a mean fill of 98.06%, against 2,928–2,931 and
 474 97.89–98.16% in runs 1–3. It costs 8.8% less than SAP-derived parameters
 475 (95% interval of the mean difference -566 to -61 ; fill -0.24 points, inter-
 476 val including zero) and 28.4–29.1% less than the SLT-informed and ERP-
 477 floor ORs, but it costs 9.0% more than the history-only OR (interval $+134$
 478 to $+387$; fill -0.09 points; cheaper in 91 of 346 pairs), 11.4% more than
 479 the Croston/SBA OR and 21.1% more than the empirical-quantile OR, and
 480 7.7% less than the simulation-tuned policy at its different operating point.
 481 Withholding the ERP parameters thus raises the LLM’s cost by roughly 3
 482 percentage points relative to the history-only OR and does not reverse any
 483 ranking.

484 *By plant.* The pattern is the same as with ERP inputs: the LLM costs
 485 less than SAP-derived at Plants A (215 pairs, -10.8%) and B (65 pairs,
 486 -21.9%) and more at Plant C (66 pairs, $+41.6\%$ against $+36.2\%$ in run 1),
 487 so the Plant C loss is not caused by reliance on ERP parameters.

488 *Interpretation.* Two conclusions follow. First, the information-
 489 asymmetry concern (that the LLM benefits from ERP inputs) does not
 490 explain the LLM’s performance: with the ERP parameters removed the
 491 LLM keeps most of its advantage over ERP-derived parameters, and the
 492 gap to the history-only OR widens only moderately. Second, the ERP
 493 parameters are of limited use to the LLM; most of its output appears to be
 494 determined by the demand and lead-time histories. One limitation is that

the ablation tests only the two ERP planning fields; the LLM still sees unit cost, holding rate, ordering cost, MOQ, service target and storage limit, which all reference arms also use.

6.5. Storage limits along simulated trajectories

All 365 LLM policies satisfy the static rule $SS+Q \leq \text{max_storage_units}$, yet in the unconstrained simulation the post-receipt on-hand stock exceeds the limit for 183 pairs (52.9%) for the LLM (178–183 across the three runs; 10.2% of simulated pair-days), 198 pairs (57.2%) for SAP-derived and 230 pairs (66.5%) for the SLT-informed OR; no pair starts above the limit. Exceedances arise because inventory on order and lead-time demand add to stock above $SS + Q$, so the static check is necessary but not sufficient for any method. Under the hard-limit replay no arm exceeds the limit by more than one unit. Enforcing it raises the LLM mean cost from 2,850 to 2,928 and SAP-derived from 3,184 to 3,305 (SLT-informed OR from 3,852 to 4,211), without changing the ranking, and we use the replay as the primary evaluation.

6.6. Plants and demand regimes

The average hides a strong plant effect (Table 8, hard-limit replay, run 1). The LLM costs 14.1% less than SAP-derived in Plant A (cheaper in 179 of 215 pairs) and 22.7% less in Plant B (54 of 65), but 36.2% *more* in Plant C, where it is cheaper in only 8 of 66 pairs. The Plant C loss is similar in the other full runs (+32.2% and +31.5%) and larger without the ERP parameters (+41.6%). Plant C is also the plant whose calendar we repaired, but Section 6.8 shows that the loss persists under every working-day reconstruction in which Plant C can be scored. Against the data-driven baselines

the LLM costs more in every plant: 7.2%, 3.6% and 6.7% more than the history-only OR in Plants A, B and C; 13.1%, 2.3% and 6.2% more than the Croston/SBA OR (it is cheaper in 26 of 65 Plant B pairs and 20 of 66 Plant C pairs); and 20.8%, 7.7% and 29.5% more than the empirical-quantile OR. Plant B is where the LLM comes closest. The SLT-informed and history-only ORs coincide in Plant C because the ERP safety lead time is zero for all 66 of its pairs.

By demand regime, the 33 lumpy materials show the largest cost reduction against SAP-derived (−25.4% in run 1; −23.7% to −25.2% in the other runs), whereas the 313 smooth materials show −9.8% at a 0.24-point lower mean fill. The service effect on lumpy materials is inconsistent (mean fill +0.38 points in run 1, −0.29 in run 2), the history-only OR costs 9% less than the LLM there at a much lower aggregate fill, and the Croston/SBA OR costs 9.7% less at a higher mean fill (94.55% versus 93.12%); only the empirical-quantile OR is slightly more expensive (1.2%); the LLM is cheaper in 15 of 33 pairs). With 33 materials this is indicative only. By terciles of the coefficient of variation, the LLM costs 15.7% less than SAP-derived at equal fill in the lowest tercile, 7.5% less with a 0.68-point lower mean fill in the middle tercile, and 12.9% less in the highest.

Table 8: Mean simulated cost per pair and mean fill by plant, run 1, hard-limit replay (common-feasible cohort). The SLT-informed and history-only ORs coincide in Plant C.

Plant (n)	LLM cost	SAP-derived cost	SLT-informed OR cost	History-only OR cost	Croston/SBA OR cost	Empirical-quantile OR cost	LLM vs SAP	Cheaper pairs	Fill LLM / SAP (%)
All (346)	2,928	3,385	4,211	2,784	2,785	2,489	−11.4%	241	98.11 / 98.20
A (215)	2,391	2,782	3,779	2,231	2,114	1,980	−14.1%	179	99.18 / 99.27
B (65)	4,916	6,369	7,229	4,768	4,896	4,566	−22.7%	54	96.23 / 96.27
C (66)	2,719	1,897	2,549	2,549	2,561	2,100	+36.2%	8	96.20 / 96.61

6.7. Run-to-run stability

Across the three runs the aggregate results are very similar: the LLM cost relative to SAP-derived is −11.4%, −11.3% and −11.4% in the hard-limit replay, and the distributions of policy values have similar medians (reorder

point 1,584, 1,582 and 1,466; order quantity 2,737, 2,777 and 2,693; safety stock 683, 688 and 669; Figure 2). Individual policies, however, vary. Comparing runs 2 and 3 on the 365 pairs, all three policy values are identical for only 36 pairs; every policy resolves to the (r, Q) family, and the median absolute differences are 81 units for the reorder point, 6 for the order quantity and 3 for the safety stock. The reorder point differs by more than 10% in about 48% of the pairs, and a few pairs vary strongly (for one material the reorder point was 14,882, 14,947 and 29,218 units in the three runs; for two others the safety stock was zero in one run and 660 or 1,075 units in the others). One generation per material is therefore a noisy estimate: aggregate behaviour is reproducible, individual values should be reviewed. Such non-determinism is a documented property of LLMs [16].

6.8. Sensitivity analyses

Plant C calendar.. Plant C’s source working-day flag is all zero, which required reconstruction. We re-scored all policies under four alternatives: any-plant, strict-all, exclusion of Plant C and calendar-day arrivals. Where Plant C can be scored (any-plant and calendar-day arrivals), the LLM remains 31–35% more expensive than SAP-derived there in the three full runs, and the cohort-wide advantage is -10.8% to -11.1% (primary calendar: -11.3% to -11.4%). The strict and exclusion variants (280 pairs) drop Plant C and show a larger advantage (-17.0% to -17.6%). The Plant C result is therefore not an artefact of the calendar repair; one residual risk is that the LLM saw Plant C demand series built on the repaired calendar.

Shortage penalty.. With a positive lost-sales penalty the zero-penalty cost ranking changes. Table 9 gives mean total cost per pair (holding, ordering and shortage cost) in the hard-limit replay for penalties of one and five times

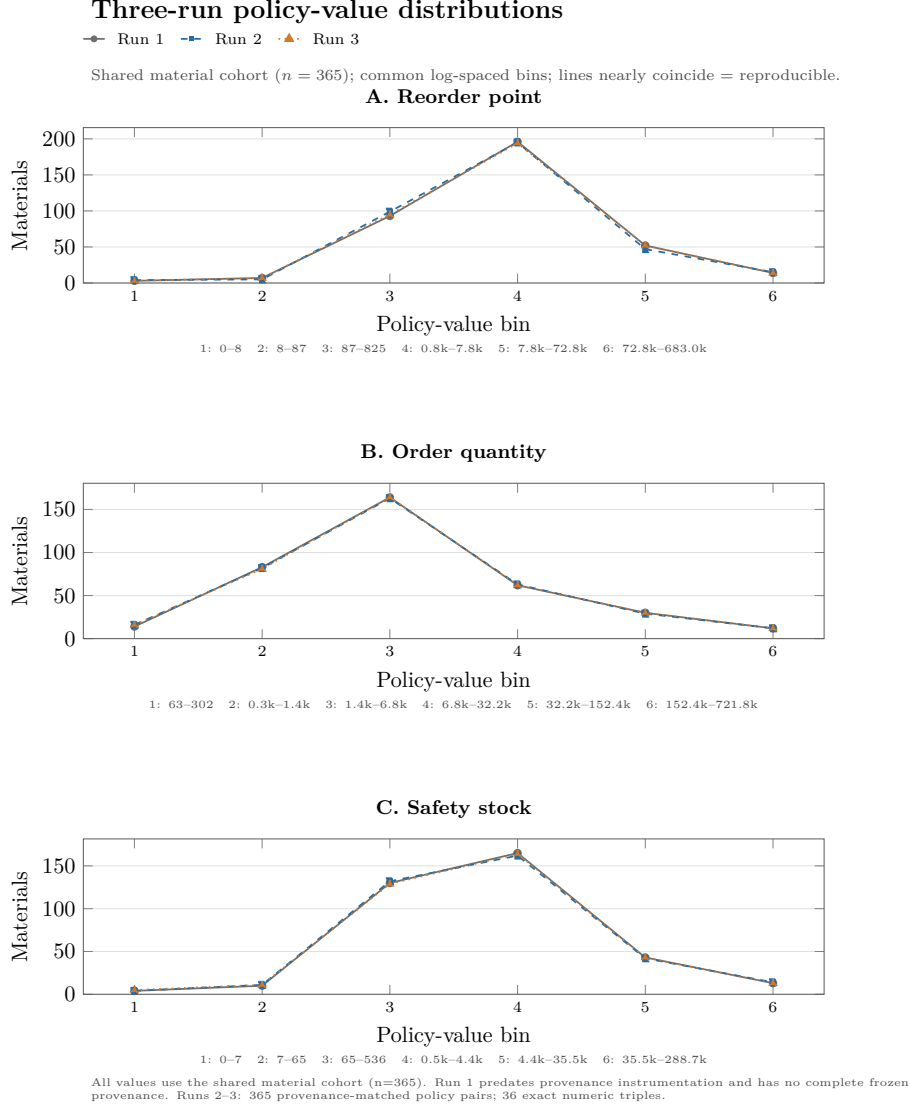


Figure 2: Policy-value distributions across the 365 shared pairs for runs 1, 2 and 3 (materials per common log-spaced bin); similar lines indicate similar aggregate distributions, not identical policies for each material.

569 the unit cost; the zero-penalty costs are in Table 7. Costs are dominated by
570 the shortage term and the gaps between arms are small in relative terms, so
571 the ordering among the non-tuned arms (differences of 0.1–3.7%) should be

read with caution. The simulation-tuned policy is by far the cheapest arm as soon as shortages are priced (1,094,043 at one times the unit cost, about 33% below SAP-derived), because it was selected to meet the service target on the pre-cutoff history. Among the other arms the SAP-derived parameters are the cheapest, because they carry more inventory and fewer stockouts. Run 1 of the LLM then costs 2.4% more than SAP-derived at both one and five times the unit cost; it is about as expensive as the Croston/SBA OR (1,673,003 versus 1,674,925 at one times the unit cost), and it is cheaper than the SLT-informed OR (1.2% lower at both multiples), the ERP-floor OR, the empirical-quantile OR and the history-only OR (0.8% lower). The LLM’s cost advantage over ERP-derived parameters is therefore specific to the case in which service shortfalls are not charged.

Table 9: Mean total cost per pair in the hard-limit replay (346 pairs) when lost sales are charged at a multiple of the unit cost.

Arm	1×	5×
SAP-derived	1,634,136	8,157,462
SLT-informed OR	1,694,183	8,454,072
ERP-floor OR	1,694,220	8,454,109
History-only OR	1,687,191	8,424,899
Croston/SBA OR	1,674,925	8,363,806
Empirical-quantile OR	1,693,784	8,458,966
Simulation-tuned (r, Q)	1,094,043	5,457,151
LLM run 1	1,673,003	8,353,306
LLM run 2	1,683,280	8,404,676
LLM run 3	1,671,759	8,347,082
LLM no ERP	1,689,961	8,437,748

7. Managerial Insights

1. **Position the tool as a reviewed decision support system, not an autonomous policy engine.** Single generations differ at the material level (Section 6.7). The gate-simulate-review structure (Section 4.3) yields a candidate policy for every material with a justification and leaves the

589 decision to the planner; whether this helps planners in practice has not
590 been tested.

591 **2. Benchmark against data-driven OR baselines before claiming**
592 **value.** ERP-derived parameters and ERP-informed OR policies are weak
593 benchmarks: safety stocks are often stale or zero and safety lead times in-
594 flate the effective lead time. A planning team that evaluates an LLM aid
595 only against ERP will overestimate its benefit. History-only (r, Q) base-
596 lines (normal approximation, Croston/SBA, empirical lead-time-demand
597 quantile) are cheap to compute and the LLM cost 6%, 8% and 18% more
598 than they did in our data, respectively; a policy tuned by simulation on
599 the pre-cutoff history was cheapest once shortages were priced.

600 **3. Where the LLM was closer to the baselines.** Against ERP-derived
601 parameters the LLM was cheaper in Plants A and B (14–23%) and on the
602 33 lumpy materials (about 25%, with mixed service effects). In Plant C,
603 where the LLM lost, conditions such as repaired calendars, small samples
604 or atypical lead-time behaviour may call for escalation; we did not test
605 this.

606 **4. Verify storage along the trajectory.** The static rule $SS + Q \leq \text{limit}$
607 cannot protect a warehouse: 53–66% of pairs breach the limit in simu-
608 lation under each of the three arms checked. Operations should enforce
609 a cap in the order-release logic (order-up-to or truncation), or treat the
610 limit as a design requirement in the policy.

611 **5. Price shortages explicitly.** Without a shortage penalty the cost rank-
612 ing rewards under-stocking. With a penalty of one unit cost the LLM’s
613 cost advantage over SAP disappears (it is about 2.4% more expensive).

614 Planners should set service targets and penalties before reading the cost
615 comparison.

616 **6. Freeze and log the configuration.** Treat the system prompt as a
617 controlled file, log every call (input hash, time, session, raw response)
618 and re-run before changing the prompt. Our reproducibility argument
619 rests on this log, not on the platform’s model identifier, which the client
620 cannot see.

621 **7. Cost and latency.** Each LLM generation took several minutes, whereas
622 the OR baseline takes milliseconds. The LLM is attractive where ana-
623 lyst time and not compute is scarce, but a deployment needs batching
624 and monitoring. Periodic regeneration as demand and lead times change
625 would also support adaptable planning under uncertainty [92].

626 **8. Discussion, Limitations and Research Perspectives**

627 *What the evidence shows..* In a three-plant industrial dataset, a prompt-
628 configured LLM generates feasible, reviewable (r, Q) policies for every ma-
629 terial. In the simulation they cost less than ERP-derived parameters but
630 more than history-only data-driven OR policies (Croston/SBA, empirical-
631 quantile). We therefore do not present InventOR as a method that outper-
632 forms optimization; its claimed role is that of a reviewed candidate generator,
633 and whether that role helps planners is a hypothesis that the simulation can-
634 not test. The information-controlled evaluation is what supports this read-
635 ing: against the ERP-informed arms alone, the LLM would appear much
636 better (-11.4% against SAP and -30.5% against the SLT-informed OR in
637 the hard-limit replay).

638 *Design principles for LLM-based decision support.* The evidence supports
639 six principles that we derive from this application and offer as hypotheses
640 for further design-science work [7]; they are not tested beyond it. **DP1,**
641 **bound the generator:** give the LLM the role of candidate generator with
642 a prompt that fixes the method, the constraints and the output schema, and
643 no authority to act. **DP2, verify deterministically and along time:**
644 check every output with code, and check feasibility along the trajectory,
645 because point checks of constraints can pass while the constraints are violated
646 over time (Section 6.5). **DP3, show counterfactual evidence:** give the
647 reviewer a simulated comparison with credible references, so that review
648 is based on outcomes and not on the persuasiveness of the justification.
649 **DP4, control information in evaluation:** compare against baselines
650 that use the same information and against a baseline that uses none, and
651 withhold incumbent inputs from the LLM in an ablation, otherwise a gain
652 may merely reflect a weak benchmark (Sections 6.3, 6.4). **DP5, freeze and**
653 **log:** treat the prompt as a controlled file, log every call and repeat the runs,
654 because individual outputs vary (Section 6.7). **DP6, escalate by rule:**
655 define in advance when proposals are trusted, escalated or rejected, to limit
656 complacency [13] (Section 4.3).

657 *Limitations.* (i) The data come from one manufacturer and three plants;
658 generalization is not shown. (ii) The platform does not expose the model
659 identifier or inference settings, so the model is operator-declared. (iii) No
660 planner outcomes were measured. The dialog component is a prototype
661 record, not an evaluated interface, and the claimed benefits of reviewability,
662 auditability and planner support are design intentions that we did not test;
663 a field study with planners is required to measure usability, workload, over-

ride behaviour and decision quality. (iv) The simulation uses a deterministic arrival at the rounded mean lead time, excludes observed receipts, corrections and scrap, and starts with an empty pipeline. Lead-time variability is not simulated, which may bias comparisons that depend on safety lead time (for example against the SLT-informed OR). We also did not report whether each arm attained the service target. (v) The primary results use a zero shortage penalty and standard holding and ordering costs; with a positive penalty the cost ranking changes. Costs are dominated by a few high-demand materials and the 346 pairs are not independent draws, so the bootstrap intervals understate uncertainty. (vi) The (r, Q) policy is a single-item, single-echelon family; multi-item coupling and shared capacity are outside the scope. (vii) Run-to-run variation of individual policies is substantial, and Plant C required calendar reconstruction. (viii) The reference arms comprise ERP-derived and normal-approximation (r, Q) policies, Croston/SBA and empirical-quantile (r, Q) policies and a simulation-tuned (r, Q) policy; we did not include deep reinforcement-learning or neural forecasting baselines, so the comparison with the state of the art is incomplete. The ERP-derived arm is the ERP safety stock combined with our own order quantity and is not an observed record of the plants' actual practice. (ix) The trust, escalation and rejection rules of Section 4.3 have not been evaluated.

Research perspectives..

1. *Ensembling and consistency.* Aggregating several generations (median or consensus policies) may reduce the individual-policy noise while retaining the LLM's justification; the copilot findings for spare parts [47] suggest this is promising.

- 690 2. *Hybrid generators.* Combining the best data-driven OR policy (Cros-
 691 ton/SBA or empirical quantile) with the LLM (the LLM reviewing and
 692 annotating the OR policy, or choosing among candidates by regime), as
 693 proposed for LLM–OR integration [93], could capture the OR’s cost level
 694 and the LLM’s flags.
- 695 3. *Comparison with trained methods.* A benchmark that compares prompt-
 696 based generators with deep reinforcement-learning and other trained poli-
 697 cies [63, 51, 52] on identical data and information would clarify the trade-
 698 off between setup effort and policy quality.
- 699 4. *Intermittent-demand and richer policy families.* Allowing order-up-to or
 700 (s, S) policies and regime-specific estimators may help the lumpy and
 701 Plant C pairs [68, 69].
- 702 5. *Planner-in-the-loop field trials.* Prospective studies should compare plan-
 703 ners working with and without the aid on a matched set of materials, and
 704 measure planner time, override rates and realized service and inventory,
 705 which our simulation cannot, and should examine adoption drivers and
 706 barriers with mixed-methods designs [29, 28, 5].
- 707 6. *Standards for information-controlled evaluation.* Studies of LLM-based
 708 decision support systems should report an ERP-independent baseline, an
 709 ablation that withholds the incumbent’s inputs, trajectory-level feasibil-
 710 ity, and run-to-run variation.

711 9. Conclusion

712 We defined InventOR as a human-reviewed decision support system for a
 713 recurring production-planning decision, the setting of replenishment parame-

ters per plant-material pair, and evaluated it against six design requirements on a three-plant industrial dataset. A prompt-configured LLM generated feasible, justified (r, Q) policies for every material without training, and a deterministic gate, a hard-limit simulation and explicit review rules made these proposals checkable by a planner. The proposals cost about 11% less than ERP-derived parameters at slightly lower service, but the LLM costs about 6% more than a history-only OR policy, 8% more than a Croston/SBA policy and 18% more than an empirical-quantile policy; when lost sales are priced at one unit cost or more, ERP-derived parameters are the cheapest of the non-tuned arms and a simulation-tuned policy is cheapest overall. Storage limits that hold in a static check are exceeded along simulated trajectories for every method, and the LLM loses to ERP at one plant. The evidence therefore supports InventOR as a checkable candidate generator inside a reviewed process, with trajectory-level storage checks, explicit shortage pricing and a frozen, logged configuration. It does not show that the LLM out-optimizes a data-driven OR baseline, and the benefit to planners remains to be tested.

The design principles (Section 8) are hypotheses for LLM-based decision support in settings where parameters are set item by item. Field evidence on planner behaviour and realized outcomes is the central open step.

Appendix A. Formal Definitions

Reference calculation. These are the standard textbook calculations [78, 87]. For pair i , let $\bar{d}_i$ and $\sigma_{D,i}$ be the mean and standard deviation of pre-cutoff working-day demand, L_i and $\sigma_{L,i}$ the mean and standard deviation of the lead time in working days, K_i the ordering cost, h_i the annual holding cost per unit (unit cost times holding rate), z_i the standard normal quantile of

739 the service target, and 250 the working days per year. The order quantity
 740 is $Q_i = \max\{\text{MOQ}_i, \lceil \sqrt{2K_i \cdot 250\bar{d}_i/h_i} \rceil\}$. The lead-time-demand standard
 741 deviation is $\sigma_{LTD,i} = \sqrt{L_i\sigma_{D,i}^2 + \bar{d}_i^2\sigma_{L,i}^2}$, the safety stock $SS_i = \lceil z_i\sigma_{LTD,i} \rceil$
 742 and the reorder point $R_i = \lceil \bar{d}_iL_i^{\text{eff}} + SS_i \rceil$, where L_i^{eff} is the mean lead time
 743 for the history-only OR and the mean lead time plus the ERP safety lead
 744 time for the SLT-informed OR and the ERP-floor OR. The SAP-derived arm
 745 uses the ERP safety stock and the same Q_i . All policies are projected to
 746 $SS_i + Q_i \leq \text{max_storage_units}_i$ when $\text{MOQ}_i + SS_i$ fits.

747 *Simulation..* On each post-cutoff working day, demand is served from on-
 748 hand stock, unmet demand is lost, and an order of size Q_i is placed when
 749 the inventory position (on-hand plus on order) is at or below R_i and ar-
 750 rives after the rounded mean lead time in subsequent working observations.
 751 Observed receipts, corrections and scrap are ignored. The total cost is
 752 $TC_i = h_i\bar{I}_iT/250 + K_iN_i$, where $\bar{I}_i$ is the mean on-hand stock over the
 753 T simulated working days and N_i the number of orders. Fill rate is served
 754 divided by demanded units. In the hard-limit replay an order is truncated
 755 to $\text{max_storage_units}_i$ minus the inventory position, and no order is placed
 756 if the truncated quantity is zero.

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768 **Data and Code Availability**

769 The raw operational data and material-level artifacts cannot be publicly
770 released because they contain commercially sensitive material, supplier
771 and planning information. A confidentiality-screened release package
772 containing analysis code, the anonymized variable schema, aggregate
773 evaluation results, manuscript sources and prompt files is available at
774 github.com/DressPD/inventor_public. A versioned Zenodo archive of
775 the release is available at <https://doi.org/10.5281/zenodo.22262052>.
776 Supplementary File S1 contains the system prompt used for all runs; the
777 prompt file, the runner code and, for each call, a hash of the input, the
778 UTC time, the session identifier and the raw response are in the release.
779 The model identifier and inference settings are not exposed by the platform
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787 **Declaration of Generative AI and AI-Assisted Technologies in the** 788 **Writing Process**

789 During manuscript preparation and revision, the author used OpenAI
790 GPT-5.6 and Anthropic Claude models through the OpenCode coding assis-
791 tant for language editing, restructuring, consistency checks, citation verifica-
792 tion against Crossref metadata, LaTeX proofing, and writing and executing
793 the analysis scripts that produce the evidence tables. These assistants are
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